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Question 1 [C03] [10 Points]

```
public class Doctor {
    public static int patients = 3, x=6;
    public int fee = 50;
    public int cases = 2;
    public Doctor() {
        fee = patients * 10;
        cases = patients + 1;
        patients += 2;
    }

    public void treat(int x, int y) {
        int bill = 0;
        cases = cases + x + patients++;
        bill = bill + y + 200;
        fee = fee + cases + bill;
        System.out.println(bill + " " + cases + " " + fee);
    }

    public int specialistFee(int y){
        return x*y+fee++;
    }
}

public class Surgeon extends Doctor {
    public static int x=6,patients = 3;
    public int operations = 4;
    public int patient = 10;
    public Surgeon() {
        operations = patients + 2;
        cases = patients - 3;
        patient -= 2;
    }

    public void treat(int x, int y) {
        int bill = 100;
        operations = operations + x;
        cases = cases + y + patients++;
        bill = bill + operations;
        fee = fee + cases + bill;
        System.out.println(operations + " " + cases + " " +
fee);
    }

    public void operate(int x, int y) {
        int operations = 1;
        operations = operations + this.operations-patient;
        this.operations = operations +
patients+specialistFee(2);
        treat(x, y);
        fee = fee + this.operations + super.cases;
        System.out.println(operations + " " +
this.operations + " " + fee);
    }
}
```

Illustrate the output of the following statements written in the main method of the driver code. [Answers without workings on the script will be rejected]

Driver code:
Surgeon s1 =new Surgeon();
s1.operate(2, 3);
Doctor d1 = new Doctor();
d1.treat(1, 2);

Output:

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Question 1 [C03] [10 Points]

```
class BankAccount {
    public static int accounts = 2, rate = 5;
    public int balance = 35;
    public int transactions = 1;
    public BankAccount() {
        balance = accounts * 5;
        transactions = accounts + 2;
        accounts += 2;
    }

    public void deposit(int amount, int bonus) {
        int total = 0;
        transactions = transactions + amount + accounts++;
        total = total + bonus + 100;
        balance = balance + transactions + total;

        System.out.println(total + " " + transactions + " "
+ balance);
    }

    public int calculateInterest(int years) {
        return rate * years + balance++;
    }
}

class PremiumAccount extends BankAccount {
    public static int accounts = 4, rate = 8;
    public int rewards = 3;
    public int transactions = 7;
    public PremiumAccount() {
        rewards = accounts + 1;
        transactions = accounts - 2;
        balance -= 100;
    }
    public void deposit(int amount, int bonus) {
        int total = 200;
        rewards = rewards + amount;
        transactions = transactions + bonus + accounts++;
        total = total + rewards;
        balance = balance + transactions + total;
        System.out.println(rewards + " " + transactions + "
" + balance);
    }

    public void purchase(int amount, int bonus) {
        int transactions = 2;
        transactions = transactions + this.transactions -
rewards;
        this.transactions = transactions + accounts +
calculateInterest(2);
        deposit(amount, bonus);
        balance = balance + this.transactions +
super.transactions;
        System.out.println(transactions + " "
+ this.transactions + " " + balance);
    }
}
```

Illustrate the output of the following statements written in the main method of the driver code. [Answers without workings on the script will be rejected]

Driver code:
PremiumAccount p1 = new
PremiumAccount();
p1.purchase(3, 2);

BankAccount b1 = new
BankAccount();
b1.deposit(1, 3);

Output:

Output for Set A

45	6	182
-2	45	233
202	14	266

Output for Set B

8	-71	48
-1	-71	-19
103	13	136

Rubric for Set A and Set B

SL	Criteria	Marks
1	Each correct output yields 1 mark	1 x 9 = 9
2	A trace table yields 1 mark	1
If no calculation/trace table is found, the student should receive 0 out of 10		